Provability facts for disjunction in tableaux

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: four derived-node formulas in this proof are written in the source's one-argument macro form (sFmla{True{...}} / sFmla{False{...}}) and render with the sentence outside the parentheses (top A () instead of top (A)); this is the author's own text delivered verbatim (left-left typo precedent); the affected lines are 1138-1139 and 1148-1149

Editorial note. The label prop:provability-lor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed tableau rules for disjunction. From True (A or B) the True-or rule branches into True A and True B; from False (A or B) the False-or rule adds both False A and False B to the same branch.. Item (1) closes via the True-or split against the derived False A and False B; item (2) closes via the False-or conclusions against the True assumptions.
- Inconsistency via closed tableaux. A set of sentences is inconsistent iff some finite subset has a closed tableau.. Item (1) is witnessed by a closed tableau for the signed assumption set {True (A or B), True (not A), True (not B)}.

Proposition 0.1. *1.* $\{A \vee B, \neg A, \neg B\}$ *is inconsistent.*

2. Both $A \vdash A \vee B$ *and* $B \vdash A \vee B$.

Proof. 1. We give a closed tableau of $\{\top(A \vee B), \top(\neg A), \top(\neg B)\}$:

1.	$\top(A \vee B)$	Ass
2.	$\top(\neg A)$	Ass
3.	$\top(\neg B)$	Ass
4.	$\bot(A)$	$\top\neg$[2]
5.	$\bot(B)$	$\top\neg$[3]
6.	$\top(A)$ $\quad$ $\top(B)$	$\top\vee$[1]
	$\otimes$ $\quad$ $\otimes$	

2. Both $\{\bot(A \vee B), \top(A)\}$ and $\{\bot(A \vee B), \top(B)\}$ have closed tableaus:

1.	$\bot(A \vee B)$	Ass
2.	$\top(A)$	Ass
3.	$\bot A()$	$\bot\vee$[1]
4.	$\bot B()$	$\bot\vee$[1]
	$\otimes$	

1.	$\bot(A \vee B)$	Ass
2.	$\top(B)$	Ass
3.	$\bot A()$	$\bot\vee$[1]
4.	$\bot B()$	$\bot\vee$[1]
	$\otimes$	

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